# DTCP Implant Power Transmitter

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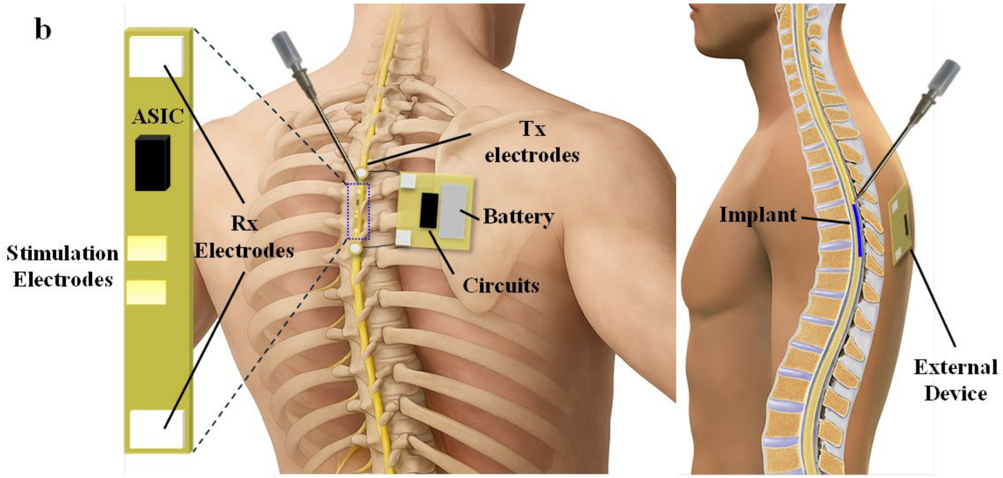
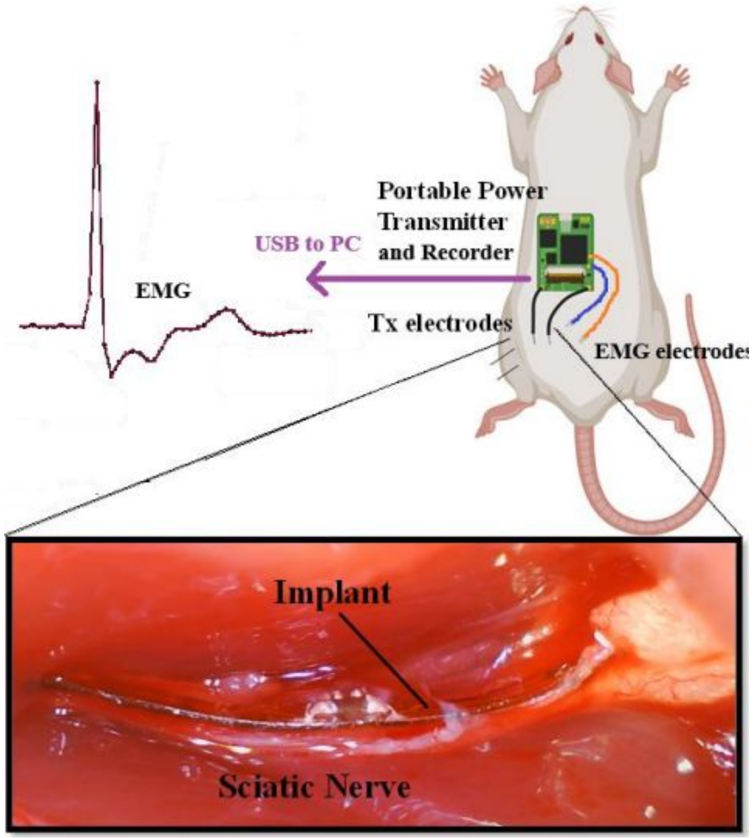
Post-mortem presentation

# Problem Statement

Objective: demonstrate a novel IMD powering technology through a proof-of-concept neural modulation use case.



DTCP system overview

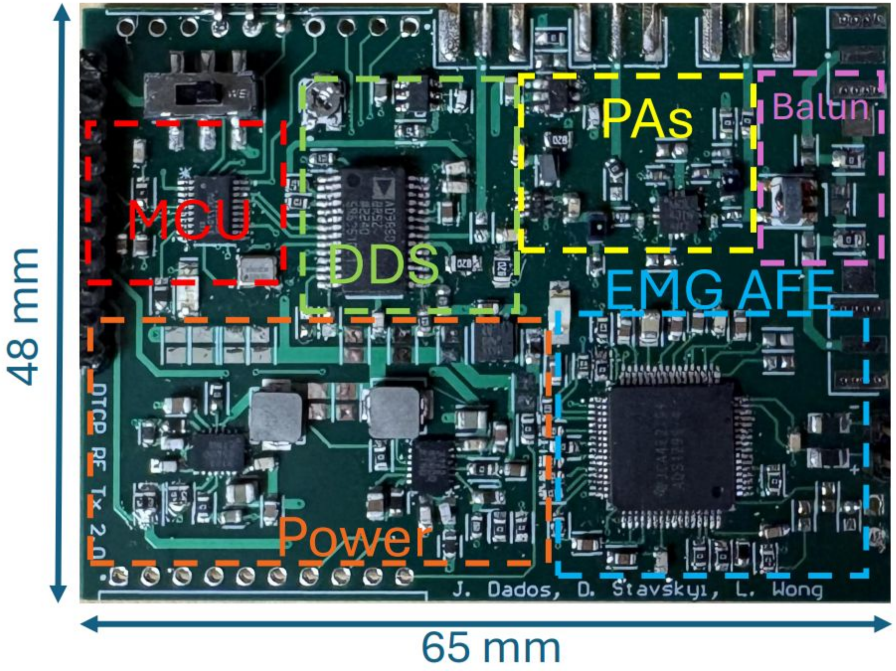


Target research application

# Project State

| Metric | Achieved value |
|---|---|
| Peak output power | 30.94 dBm |
| Peak current consumption (during power transmission) | 1.35 A |
| Average current consumption (during EMG recording) | 20 mA |
| EMG sampling rate | 8 kS/s |
| Device dimensions | 48 mm x 65 mm |

Performance metrics




The final device

# Results (RF)

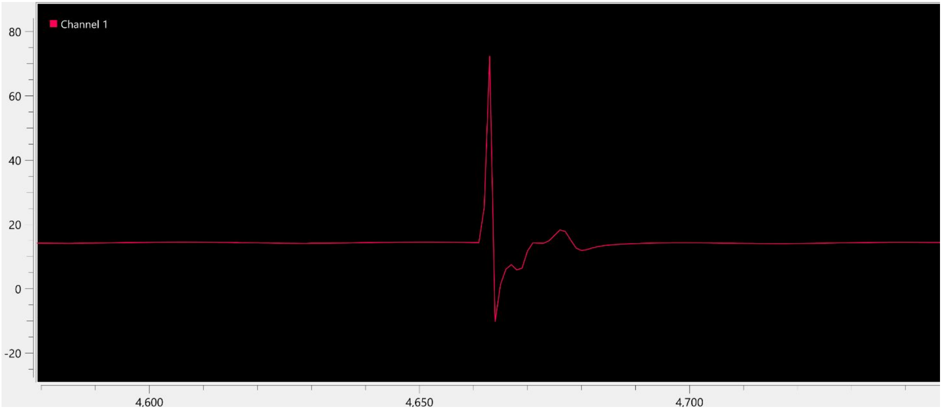
The peak output power was validated using a spectrum analyzer.


Spectrum analyzer reading (with a 10dB attenuator)

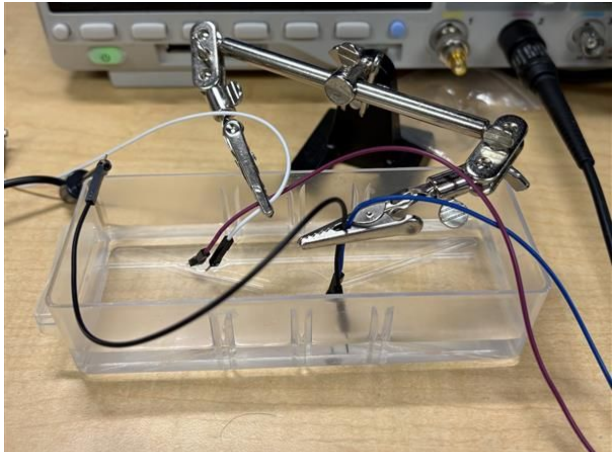

Peak power consumption

# Results (EMG)

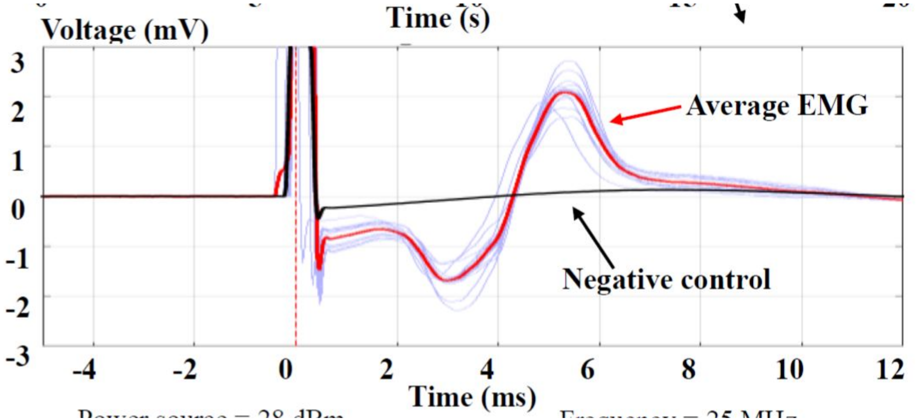
The device can replace the previous benchtop setup used at the lab.




EMG signal acquired with our device

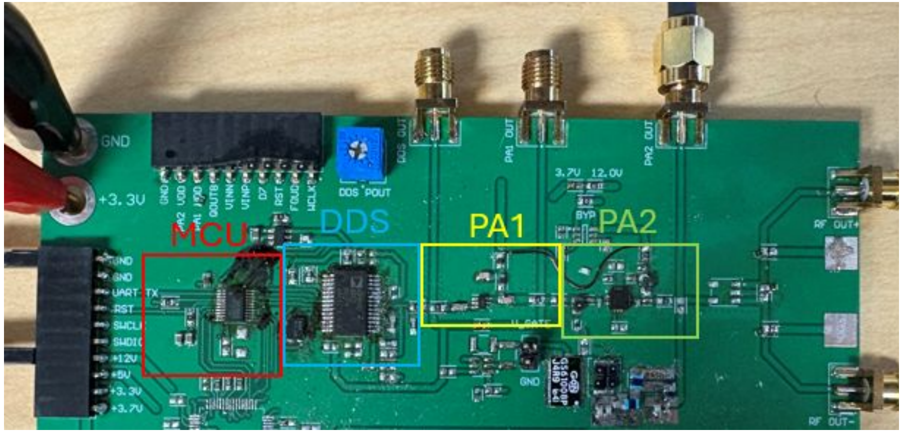


PBS setup

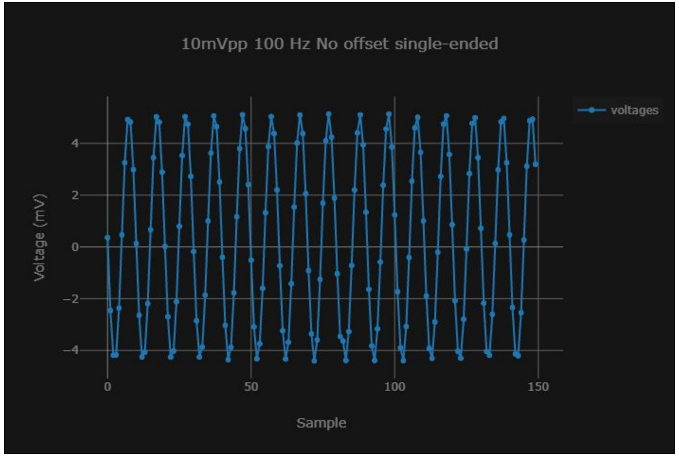



EMG signal acquired by a benchtop setup at the lab

# Successes



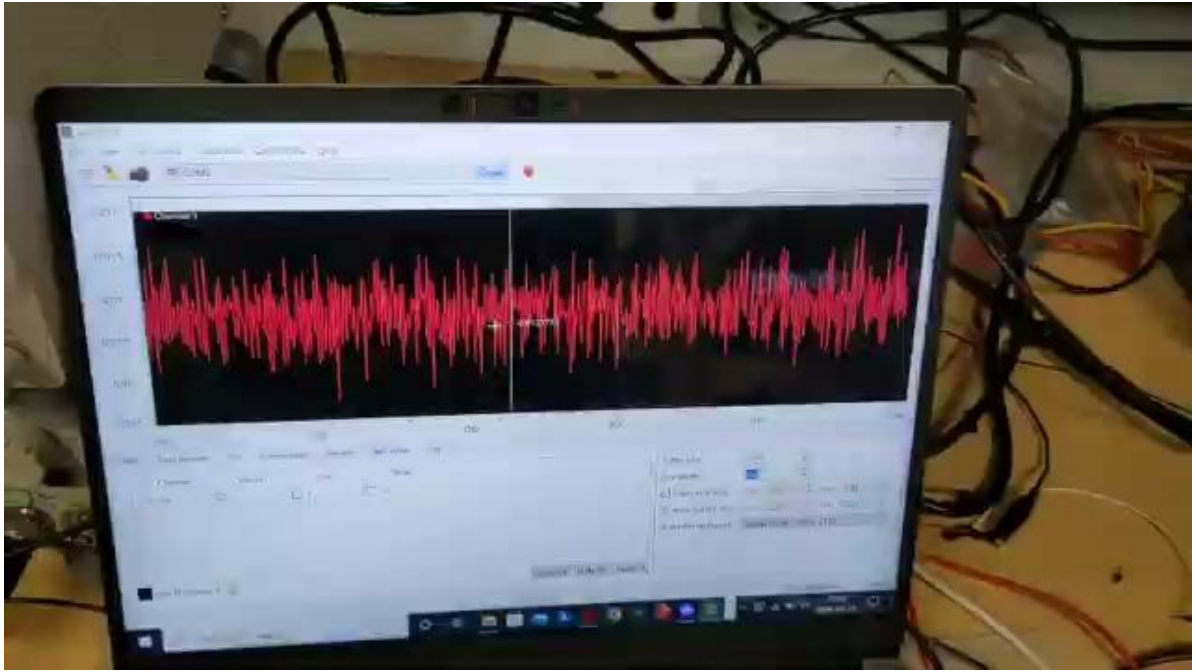

De-risking a novel design by building a modular, large PCB first




Obtaining the first measurements using our AFE

# Challenges


Soldering tiny components by hand



Grounding and bias issues making the circuit behave in unexpected ways

Thank you very much!